Hackathon

Development of Scalable, Efficient, and Indigenous Electronics Solutions Across Mobility, Power, Robotics, and Manufacturing.

Jointly hosted by
Ratan Tata Innovation Hub (RTIH)

Background & Context

India’s industrial electronics sector is undergoing a transformative shift driven by electrification, automation, and the need for indigenous innovation. From electric mobility and power electronics to industrial robotics and electronics manufacturing, the demand for scalable, efficient, and locally developed solutions is surging. However, challenges persist: reliance on rare earth materials, fragmented manufacturing workflows, and limited AI integration in automation systems.

This challenge seeks to catalyze innovation across segments, namely Mobility, Power Electronics, Industrial Automation& Robotics, and Electronics System Design by inviting solutions that are affordable, scalable, and aligned with India’s strategic priorities..

Challenge Overview

The Hackathon invites startups, innovators, institutions, and technology providers to help catalyze India’s leadership in industrial electronics innovation by providing design and develop solutions that are:

- Indigenous and Sovereign: Reducing reliance on imported systems and rare-earth materials by nurturing homegrown technologies.
- Scalable and Efficient: Enabling deployment across diverse industrial and mobility applications, from EV powertrains to collaborative robotics.
- Affordable and Accessible: Ensuring cost-effectiveness and rapid adoption across India’s manufacturing, energy, and construction ecosystems.
- Future-Ready: Integrating AI, modular architectures, and advanced design methodologies to prepare India’s industries for global competitiveness.

I. Challenge Vision and Rationale

Overarching Goal

To develop solutions that combine technical rigor, cost-effectiveness, and real-world applicability to

- Strengthen India's manufacturing and energy infrastructure; enhance safety, efficiency, and automation across construction, logistics, and industrial domains
- Create a robust pipeline of indigenous technologies ready for commercialization and large-scale deployment
- Position India as a global hub for industrial electronics innovation, delivering strong domestic impact and international competitiveness

Rationale

India's industrial electronics sector is at a critical inflection point. The convergence of **electrification, automation, and digital intelligence** is reshaping industries, yet several systemic challenges persist:

- **Rare-Earth Dependency**: Current reliance on imported permanent magnet motors and materials poses risks to supply chain resilience and cost stability.
- **Fragmented Manufacturing**: Electronics manufacturing workflows remain inconsistent, limiting scalability and quality assurance.
- **Limited AI Integration**: Automation systems often lack advanced AI capabilities, reducing efficiency and adaptability in real-world deployments.
- **High Costs of Adoption**: Industrial robotics and power electronics solutions frequently face barriers to affordability and ROI, slowing adoption in critical sectors.

II. Targeted Participants and Eligibility

The challenge is open to participants across the construction and robotics ecosystem, including:

Category	Description
Startups (Early to Growth Stage)	Industrial electronics, robotics, and power startups with scalable solutions
Independent Innovators	Individual developers with functional prototypes or algorithms
Academic & Research Institutions	Universities, labs, and incubators working on electronics and automation technologies
Technology Companies	Firms with existing platforms or components ready for customization and deployment

III. Scope of Solution

Participants are expected to design solutions covering any of the following functional scope:

Mobility

- Design and implement an Advanced Driver Assistance Systems (ADAS) vision solution leveraging Vision Transformers (ViT) that achieves high accuracy, robustness, and scalability while operating under low-compute constraints suitable for real-world automotive deployment.
- Design and develop a rare earth–free electric powertrain (motor and motor control unit) for 3W, 4W, and commercial vehicles (Buses/Trucks) that matches or exceeds the performance and efficiency of conventional PMSM-based systems while ensuring scalability, cost-effectiveness, and real-world automotive viability

Power Electronics

- Design and develop a high-efficiency, modular 40 kW AC–DC converter for EV charger applications with scalable power capability (30 kW–60 kW), innovative architecture, and performance benchmarking, while achieving a bill of materials (BoM) cost below $350 at a production volume of 10,000 units.
- Design and demonstrate a Battery Management System (BMS) for utility-scale Battery Energy Storage Systems (BESS) that delivers high-fidelity state estimation, robust safety response, efficient cell balancing, and readiness for grid support and integration.

Industrial Robotics & Automation

- Prototype a collaborative robot (COBOT) for tile laying and solar panel installation that improves productivity and quality while achieving a return on investment (ROI) within 6–8 months through cost-effective design, ease of deployment, and operational efficiency.

Electronics System Design & Manufacturing

- Design and develop an ML-assisted Automated Optical Inspection (AOI) system for electronics manufacturing that accurately classifies defects in alignment with IPC A-610 standards, provides human-readable rationale for each decision, and improves inspection accuracy, consistency, and manufacturability.

IV. Evaluation and Selection Criteria

Proposals will be evaluated by an expert committee based on the following criteria:

Evaluation Parameter	Weightage
Innovation & Originality	20
Technical Completeness & Scalability	10
User Experience	20
Real-World Applicability & Cost Effectiveness	40
Documentation & Demo Quality	10
Total	**100**

V. Rewards & Adoption Pathway

VI. Expected Deliverables

Participants must submit:

- Working prototype or simulation
- Source code and setup documentation
- Use-case demonstration aligned to problem statement
- System architecture diagram
- Demo video or live presentation.

VII. Goal & Expected Impact

Aims to:

- Accelerate indigenous innovation in industrial electronics
- Reduce reliance on rare earths and imported systems
- Improve safety, efficiency, and automation across sectors
- Strengthen India’s manufacturing and energy infrastructure.